

MEIRINGEN, Switzerland

WMO: 066140

Lat: 46.743N

Lon: 8.110E

Elev: 579

StdP: 94.56

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -11.0	(c) -8.8	(d) -13.2	(e) 1.3	(f) -10.6	(g) -11.0	(h) 1.6	(i) -7.1	(j) 10.2	(k) 1.7	(l) 8.2	(m) 3.6	(n) 2.1	(o) 100	(p) 0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.2	(c) 28.8	(d) 19.9	(e) 26.8	(f) 18.9	(g) 24.9	(h) 18.0	(i) 20.5	(j) 26.9	(k) 19.5	(l) 25.3	(m) 18.6	(n) 23.9	(o) 3.1	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	14.0	23.8	17.2	13.2	22.5	16.4	12.5	21.3	61.7	26.7	57.9	25.3	55.0	23.8	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.5	(b) 6.7	(c) 5.4	(e) -14.7	(f) 31.7	(g) 2.6	(h) 1.5	(i) -16.6	(j) 32.7	(k) -18.1	(l) 33.6	(m) -19.6	(n) 34.4	(o) -21.5	(p) 35.5		
(4) 8.5	(5) 6.7	(6) 5.4	(7) -14.9	(8) 22.6	(9) 2.4	(10) 1.5	(11) -16.7	(12) 23.6	(13) -18.1	(14) 24.4	(15) -19.5	(16) 25.3	(17) -21.2	(18) 26.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-0.7	0.4	4.8	9.4	12.4	16.9	18.7	17.7	14.2	10.2	4.1	0.8	(6)
(7)		DBStd	7.66	4.00	4.06	3.69	3.83	3.78	3.44	3.01	3.25	2.98	3.38	4.11	3.60	(7)
(8)		HDD10.0	1348	331	269	163	57	20	1	0	0	3	38	181	286	(8)
(9)		HDD18.3	3464	589	502	419	269	186	67	31	51	129	252	427	543	(9)
(10)		CDD10.0	1029	0	0	2	37	95	208	269	241	127	45	4	1	(10)
(11)		CDD18.3	104	0	0	0	0	2	24	42	33	4	0	0	0	(11)
(12)		CDH23.3	741	0	0	0	2	27	188	319	193	12	0	0	0	(12)
(13)		CDH26.7	161	0	0	0	0	3	40	78	39	0	0	0	0	(13)
(14)	Wind	WSAvg	1.8	1.6	2.0	2.4	2.3	2.0	1.9	1.6	1.6	1.4	1.6	1.5	(14)	
(15)	Precipitation	PrecAvg	1391	90	94	92	100	143	144	149	150	111	98	115	106	(15)
(16)		PrecMax	1825	245	296	183	200	251	231	286	309	204	184	345	308	(16)
(17)		PrecMin	1012	8	20	23	26	39	76	74	41	51	2	16	0	(17)
(18)		PrecStd	163	59	64	48	38	63	39	49	54	39	45	69	65	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	12.8	17.8	23.4	26.0	30.1	31.1	30.0	24.8	20.6	17.2	11.9	(19)
(20)		MCWB	4.9	5.4	9.7	13.6	17.0	20.1	20.8	20.4	16.6	12.2	10.0	6.3	(20)	
(21)		2%	DB	7.0	9.5	15.2	21.1	23.2	27.8	29.0	27.6	22.9	18.2	13.9	9.3	(21)
(22)		MCWB	4.3	4.9	8.4	12.7	15.5	19.0	19.9	20.0	16.4	11.8	8.7	5.3	(22)	
(23)		5%	DB	5.4	7.7	13.5	19.0	21.6	25.2	26.9	25.2	21.0	16.6	11.9	7.4	(23)
(24)		MCWB	3.5	3.6	7.6	11.3	14.6	18.1	18.9	18.6	15.5	11.6	8.1	4.3	(24)	
(25)		10%	DB	4.1	5.9	11.2	16.8	19.5	23.5	25.0	23.5	19.3	15.2	10.1	5.9	(25)
(26)		MCWB	2.7	3.2	6.5	10.0	13.9	17.2	18.2	17.8	14.8	11.4	7.3	3.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	6.6	10.2	14.6	17.8	21.2	22.2	21.7	18.5	15.5	11.0	7.3	(27)
(28)		MCDWB	7.4	10.3	15.9	22.1	24.3	27.9	30.2	28.3	23.3	17.8	15.0	9.4	(28)	
(29)		2%	WB	4.7	5.2	9.1	13.2	16.4	19.7	20.5	20.4	17.2	14.1	9.7	6.1	(29)
(30)		MCDWB	6.1	8.4	14.9	20.3	22.5	26.1	27.1	26.5	21.3	16.2	13.2	8.5	(30)	
(31)		5%	WB	3.7	4.2	8.0	11.8	15.1	18.7	19.5	19.4	16.2	13.0	8.8	5.1	(31)
(32)		MCDWB	5.0	6.7	12.8	17.6	20.2	24.5	25.4	24.4	20.1	14.8	11.4	7.0	(32)	
(33)		10%	WB	2.8	3.3	6.9	10.5	14.1	17.6	18.6	18.3	15.2	12.0	7.7	4.0	(33)
(34)		MCDWB	4.0	5.5	10.4	15.8	18.7	22.9	24.0	22.9	18.7	14.4	9.6	5.4	(34)	
(35)	Mean Daily Temperature Range	MDBR	5.6	7.3	9.1	10.3	9.8	9.8	10.2	9.4	8.6	7.6	6.2	6.2	(35)	
(36)		5% DB	MCDBR	7.3	10.3	13.3	15.2	14.1	13.1	13.7	12.2	11.1	10.2	9.0	8.6	(36)
(37)		MCWBR	4.8	6.3	7.7	8.0	7.6	7.2	6.8	6.5	6.4	5.7	5.1	5.4	(37)	
(38)		5% WB	MCDBR	6.1	9.4	12.4	14.0	12.7	12.2	12.8	12.0	10.0	8.0	8.3	6.9	(38)
(39)		MCWBR	4.5	6.0	7.5	7.8	7.1	6.9	6.9	6.6	6.1	5.2	5.3	5.3	4.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.288	0.321	0.376	0.428	0.450	0.454	0.455	0.439	0.405	0.369	0.322	0.285	(40)	
(41)		taud	2.299	2.197	2.100	2.049	2.076	2.131	2.133	2.188	2.261	2.324	2.356	2.323	(41)	
(42)		Ebn at Noon	797	834	833	823	819	817	809	806	800	773	747	755	(42)	
(43)		Edh at Noon	76	104	135	156	158	150	148	135	114	92	73	67	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.42	2.33	3.52	4.41	4.89	5.75	5.67	4.93	3.82	2.49	1.46	1.17	(44)	
(45)		RadStd	0.17	0.30	0.51	0.67	0.47	0.53	0.50	0.45	0.36	0.27	0.20	0.17	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46) N/A	
(47)	Regional (119 neighbors)	N/A	N/A	N/A	+0.73	+0.41	+0.32	N/A	N/A	+58	+16	(47)

Nomenclature: See separate page